A Randomized Experiment of a Schema-Based Instructional Approach to Proportional Problem-Solving for Seventh Grade Students

UNIVERSITY OF MINNESOTA
Driven to Discover℠

Asha K. Jitendra, Michael R. Harwell, Danielle N. Dupuis, Stacy R. Karl, Amy E. Lein, Susan C. Slater, Gregory Simonson, & Michael Herriges

Abstract

This study examined the effect of schema-based instruction (SBI) on seventh-grade students' mathematical problem solving performance. Using a pretest—posttest—retention test design, the study compared the learning outcomes for 1,999 students in 82 classrooms who were randomly assigned to treatment (SBI) or control condition. Results of multilevel modeling indicated significant differences favoring the SBI intervention on proportional problem solving at the posttest and retention test. There were no between-group differences on the Process and Applications subtest of the GMADE. Statistically significant differences favored the treatment group on the metacognition assessment. These results demonstrate that SBI can be more effective than the control approach in improving students' proportional problem solving.

Introduction

Conventional efforts to improve the teaching and learning of proportions have had limited success, as evidenced by reports showing that college students and young adults continue to struggle with proportional reasoning (Ahl, Moore, & Dixon, 1992; Cai & Wang, 2006; Fujimura, 2001; Jitendra et al., 2011a; Lamon, 2007). The purpose of the present study was to evaluate the effectiveness of SBI, which has shown promise in prior work in developing students' understanding of the underlying mathematical structure of proportion problems and enhancing their ability to solve problems (e.g., Jitendra et al., 2009, 2011b, 2013). SBI draws upon schema theory and is designed to help students more effectively categorize problems by emphasizing the underlying mathematical problem structure; visual representations in the form of schematic representations are used to aid in problem categorization (see Marshall, 1995). In the current study, we were interested in determining whether a similar implementation of SBI, as in Jitendra et al. (2013), would have comparable effects when we increased the geographic and socioeconomic status (SES) diversity of schools, the number of teachers and students involved, and eliminated the direct involvement of the research team in supporting both SBI classroom implementation and test administration.

Method

Participants. The sample consisted of seventh-grade students (N = 1,981) and their teachers from 82 classrooms at 58 middle schools in 50 public school districts in an upper Midwest state.

Design and Procedures. We used a randomized pretest—posttest—retention test design. The 82 classrooms were randomly assigned to SBI (j = 40) or control (j = 42). The PPS, GMADE, and metacognition assessments were group-administered by the teachers following standardized protocols. All student measures were administered at pretest and posttest with pretests given in December of the school year. Treatment teachers participated in 16 hours of professional development in mid-December. The SBI intervention was conducted five days a week across six weeks between January and February. Teachers supplanted the lessons on ratio/proportion and percent in their curriculum with the SBI lessons. In the same time period, students in the control condition were taught the same topics using their district-adopted curriculum. Students were administered the posttests during the week following the last day of the intervention, with the PPS test also administered nine weeks later.

Intervention. In treatment classrooms, SBI was implemented in five 45-50 min mathematics classes a week over 6 weeks. In the same time period, students in the control condition were taught the same topics (e.g., rates/proportions, scale drawings, percents) using instructional practices specified in their textbook. Within SBI, teachers used the following four instructional practices of (a) priming the mathematical structure of problems, (b) visually mapping information in the problem using schematic diagrams, (c) providing explicit instruction on a problem-solving heuristic, and (d) developing procedural flexibility.

Fidelity of implementation. Teachers' mean total score across the seven items on the fidelity checklist was 16.85 (SD = 2.83) out of a possible 21 points, suggesting that treatment teachers implemented SBI with generally satisfactory levels of integrity. Correlations between treatment fidelity and PPS pretest-posttest and pretest-delayed posttest gains were 0.43 and 0.20, respectively. These results suggest that teachers who showed greater treatment fidelity were associated with students showing greater gains in mathematical problem solving.

Measures. The PPS test includes 23 multiple-choice items and four short-answer items, which assess students' knowledge of ratio, proportion, and percent concepts and proportional reasoning. Omega for the PPS pretest, posttest, and retention test was .69, .77, and .76, respectively. We used the Process and Applications subscale of the GMADE, which is a 30 item test that measures students' ability to comprehend mathematical language and concepts and apply relevant operations to solve word problems across multiple content areas. Omega for the GMADE pretest and posttest were .61 and .69, respectively. We assessed students' metacognition in proportional problem solving using a researcher-developed test. This test presents a proportion word problem that has an adequate level of difficulty that cannot be solved so that students are required to use a varied set of metacognitive activities as they are asked to read eight pairs of statements and select one of each pair of statements that best describes how they would think about and work to solve similar types of problems. Omega for the metacognition pretest and posttest was 0.36 and 0.42, respectively.

Results

Regarding the PPS test, results indicated that treatment was a statistically significant predictor of PPS posttest scores, with SBI classrooms outperforming control classrooms on average, $\gamma 01$ = 1.70, $t(72)$ = 5.38, p < .001, d = 0.46. The proportion of the classroom intercept variance explained by the addition of the treatment variable above and beyond that attributable to other predictors in the model was 35%. Similarly, results for the PPS retention test indicated that the treatment variable was a statistically significant predictor, with SBI classrooms outperforming control classrooms, $\gamma 01$ = 1.22, $t(70)$ = 4.17, p < .001, d = 0.32. Treatment accounted for 32% of the between-classroom intercept variance above and beyond that attributable to other predictors in the model. Regarding the GMADE, results indicated that the treatment variable was not a significant predictor of students' GMADE posttest scores. For the metacognition posttest, the treatment variable was a statistically significant predictor of metacognition posttest scores, γ_{01} = 0.48, $t(72)$ = 4.55, p < .001, d = 0.31, and the proportion of between-classroom intercept variance attributable to this effect 30%.

SBI – Examples of Problem Types

Problem Type	*Example of Problem*
Ratio 	On Thursday, the cafeteria at Osseo Middle School sold: 42 smoothies, 75 main line lunches, 80 cookies, 51 bags of chips, 100 salad bar lunches, and 26 breakfast bars. What is the ratio of the number of main line lunches sold to the number of salad bar lunches sold on Thursday?
Percent: Part-whole comparison 	On a chapter test, Janie got a grade of 80%. The test had a total of 35 possible points. How many points did Janie earn on the test?
Percent of change 	Mariah and Alex both started exercising more and each lost 8 pounds. Before they started exercising, Mariah weighed 160 lbs. and Alex weighed 200 lbs. Who had the greater percent of decrease in weight, Mariah or Alex?
Proportion 	The Frank family from Minnesota, USA, is going to Britain for their summer vacation. They exchanged $50 for 27 British pounds. At that exchange rate, how many British pounds could they get for $75?
Scale Drawing 	On the map of the Disney World Theme Parks, a scale of 1 inch represents 3 miles. What is the actual distance between Magic Kingdom and Animal Kingdom when it is 4 inches on the map?
Percent of change 	Alan's parents increased his weekly allowance by 10% from last year. If he was getting $20 a week last year, how much will he get per week this year?

Conclusion

The results of our study corroborate the view that SBI enhances problem solving involving ratio, proportion, and percent and was found to be more effective than regular mathematics instruction. First, treatment classes outperformed comparison classes on a measure of proportional problem solving. Second, relative to control students, SBI students maintained their superior performance on the follow-up posttest. Third, SBI intervention had a significant, positive effect on metacognition scores relative to the control group. While SBI explicitly interweaves metacognition with problem solving strategies, metacognition was not explicitly addressed in the control curricula or instruction implemented in the control classrooms.

Authors' Note. The research reported here was supported by the Institute of Education Sciences, U.S. Department of Education, through Grant #R305A110358.